



Society of Petroleum Engineers

SPE-226879-MS

Multi-Engine AI Tool for Oil Field Development Plan Optimization

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226879-MS](https://doi.org/10.2118/226879-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Amount of mature fields has never been higher than today. Our objective was to develop a tool capable of fast generation of an optimized scenario of field development. Multivariate optimization workflow enhanced by AI algorithms, offers a wide assessment of field data. It is capable of proposing the most promising well control parameters and suggesting the most effective type of workovers and possible ways of well layout modifications. The input for the tool includes a very few and easy-to-obtain production parameters such as production and injection rates, pressure, etc. The tool is denoted as "multi-engine" because it uses several predictive models in the optimization workflow to gather various insights about the field from the production data and obtain the results. For example, material balance model in a form of Capacitance-Resistance model (CRM) is used for interwell connectivity field identification and enables the infill drilling optimization. As for engines, physics-driven (a modified CRM), data-driven (Recurrent Neural Network, or RNN) and hybrid (Physics-Informed Neural Network) models are used. The tool is designed to gain the computational benefits from both data-driven and physics driven models.

Methods and results

The tasks of optimizing well control and monitoring their efficiency to achieve production targets are among the most critical in oil and gas field development and operation. An analysis of relevant business processes across various oil and gas companies revealed that solving such tasks requires significant computational and time resources. In addition to handling numerous disparate files, multiple software tools are often needed to address a single task. The most time-consuming stage involves using hydrodynamic simulators, where running a single scenario for an adapted model can take from one hour, depending on the model's complexity and the number of wells. Implementing a tool capable of handling optimization and monitoring tasks end-to-end, including raw data processing, would significantly accelerate decision-making and improve field development efficiency. This paper presents the experience of developing such a solution based on machine and deep learning algorithms.

The cornerstone of the proposed system is a module that generates predictive analytics for total fluid, oil, and water production rates. Notably, this module offers users a choice of several predictive models: an analytical model (based on CRM, Capacitance-Resistance Model) [1], a neural network model (based on

RNN, Recurrent Neural Network), and a hybrid model (based on PINN, Physics-Informed Neural Network) [2]. Different fields, depending on data quantity and quality, fluid types, etc., may be better described by specific model types. To obtain production forecasts, users must upload files containing a limited set of field data: historical production rates, historical and planned injection volumes, two historical and planned bottomhole pressure values, a list of performed well interventions, and relative reservoir coordinates to determine well placement. These data are used to configure (train) the models and generate forecasts for planned operating modes. The trained models are subsequently employed to solve inverse optimization problems. The basic architecture of the solution is shown in Figure 1.

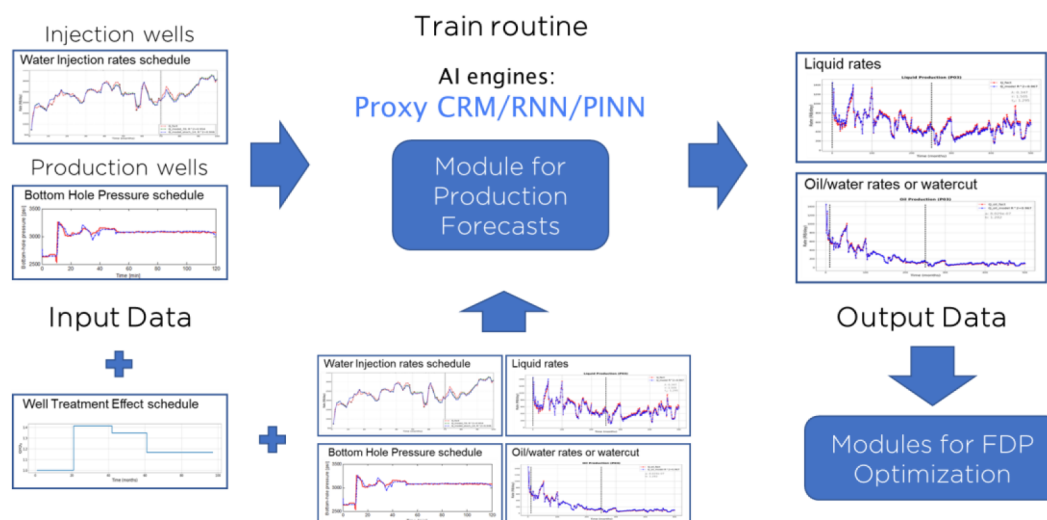


Figure 1—The architecture of the proposed solution. FDP – field development plan.

The Proxy-CRM engine is based on advanced CRM models and is capable of working with production histories, including intensive workovers, throughout the lifetime of a field. The functionality contains

- Fast short-term and middle-term production forecast for given well operation processes and layout,
- Taking into account the switch-offs of wells and the transition from production wells to injection wells,
- Accounting for new wells and neighbor wells,
- Good results for two-phase systems (oil and water) and gas single-phase systems,
- Possibility to work with wellhead pressure instead of bottomhole pressure,
- Accounting for multilayer reservoirs and well completion systems.

CRM is a data-driven, material-balance-based flow model that captures transmissibility trends, accounts for sealing or leaking faults with compressibility effects, and quantifies dissipation between injector-producer pairs. It features adjustable time constants and interwell connectivity parameters. To train the model, parameters are first initialized using analytical or random approximations and then refined through an optimization procedure. Historical-based Capacitance Models are particularly useful for poorly studied fields, enabling rapid optimization of field development strategies by simulating various well exploitation scenarios. These models require only historical data—such as hydrocarbon production volumes, injection profiles, and bottom-hole pressure dynamics—as input. A key limitation of such models is their approximate treatment of interwell properties, which are assumed constant over the entire development history. This assumption becomes problematic in cases involving workover operations or well shut-ins, as it fails to account for dynamic changes in reservoir behavior. The Proxy-CRM overcomes these issues. In this

model, we introduced mask-based indicator functions that allow deviations (new learning coefficients) to be incorporated into optimised reservoir characteristics during geological and technical operations. At the same time, other well parameters are dynamically reweighted in real time to account for the system's overall response. This parameter-updating procedure can be applied as frequently as historical data permits. However, to maintain computational efficiency, the number of new optimization parameters was kept to a minimum. Several synthetic models featuring heterogeneities and well treatments were simulated using a hydrodynamic simulator. Predictions from the enhanced CRM were compared against simulator results, with high coefficients of determination confirming forecast accuracy. While recalculated parameters incorporate corrections only for the treated well, the predicted total production rates align with target values. This demonstrates our method's ability to quantify how interventions on one well influence the performance of others.

The RNN engine is a flexible deep learning data-driven model. The functionality contains

- Fast short-term and middle-term production forecast for given well operation processes and layout,
- Accounting for well completion systems and neighbor injection wells,
- Good results for three-phase systems (oil, water and gas),
- Accounting for geology,
- Customisation for a specific type of reservoir (unconventional resources, underground storage systems, etc.).

Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures, have become valuable tools for predicting oil and water flow rates in reservoir engineering due to their ability to capture complex temporal patterns in production data. These models process sequential inputs like bottom-hole pressures, injection rates, and historical production data to forecast future fluid rates, offering advantages over traditional methods by accounting for dynamic reservoir changes and enabling real-time optimization. Typical implementations involve preprocessing time-series well data, training LSTM networks with attention mechanisms for improved interpretability, and validating against numerical simulations or field measurements. While RNNs provide fast predictions useful for production optimization and anomaly detection, challenges include sensitivity to data quality and potential physics violations, leading to growing interest in hybrid physics-informed architectures. Recent advancements explore transformer models for longer sequences and reinforcement learning integration for adaptive control, positioning RNNs as powerful surrogates that complement conventional reservoir simulation while enabling faster decision-making in field management scenarios. The technology continues evolving with techniques like transfer learning for data-scarce applications and uncertainty quantification methods to improve reliability, making these data-driven approaches increasingly practical for reservoir engineers seeking to balance computational efficiency with predictive accuracy. Our RNN engine efficiently predicts oil and water flow rates for production optimization, using LSTM cells to capture long-term dependencies in sequential bottom-hole pressure (BHP) and injection rate data. The model is trained on more than 10 oil fields real data covering diverse injection and production scenarios.

The PINN engine combines material balance and deep learning data-driven approaches to improve the model's flexibility while maintaining its physical nature. The functionality contains

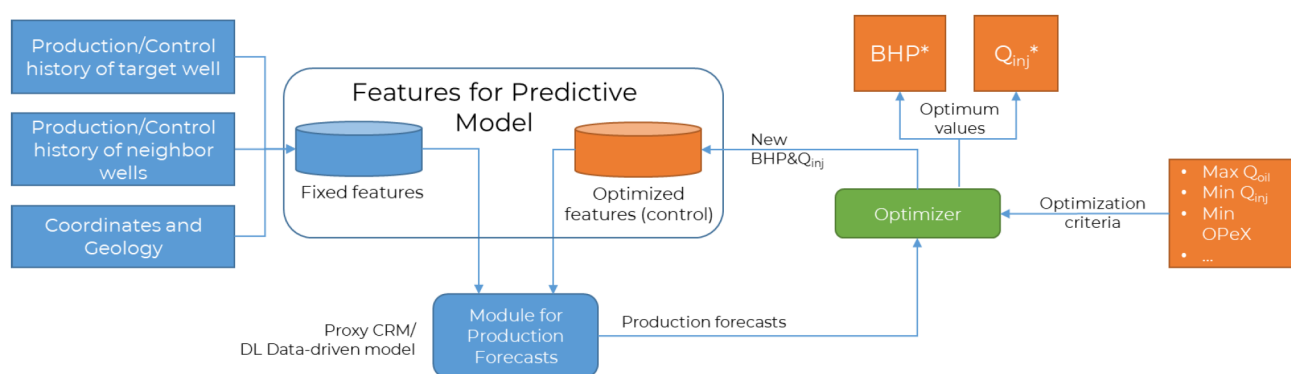
- Fast short-term and middle-term production forecast for given well operation processes and layout,
- Accounting for well completion systems and neighbor wells,
- Good results for three-phase systems (oil, water and gas),
- Low requirements for data quality.

Physics-Informed Neural Networks (PINNs) have emerged as a powerful hybrid approach for predicting oil and water flow rates by integrating data-driven machine learning with fundamental physical principles governing reservoir behavior. Unlike purely data-driven models (LSTM, RNN), PINNs incorporate governing equations (e.g., mass conservation, Darcy's law) directly into the neural network's loss function, ensuring predictions adhere to physical constraints while maintaining the flexibility to learn from observed production data. This architecture typically combines measured field data (e.g., bottom-hole pressures, production rates) with physics-based residuals, enabling accurate forecasting even in scenarios with sparse or noisy training data. For multiphase flow prediction, PINNs can encode the Buckley-Leverett equation or two-phase flow differential equations, providing more reliable estimates of water breakthrough and production decline compared to conventional machine learning models. Key advantages include improved generalization to unseen operational conditions, reduced need for extensive training data, and inherent physical interpretability—critical for reservoir engineering decisions. However, challenges remain in handling complex heterogeneities, scaling to large reservoir models, and balancing the weighting between data-fit and physics-based loss terms. Recent advances address these limitations through adaptive loss weighting schemes, hybrid architectures coupling PINNs with traditional numerical simulators, and domain decomposition techniques for large-scale applications. When properly implemented, PINNs offer a promising paradigm for predictive reservoir management, enabling faster optimization of production strategies while maintaining physical consistency—particularly valuable for mature fields with historical production data or newly developed assets with limited measurements but well-understood physics. Future directions may focus on real-time updating schemes for continuous learning and integration with uncertainty quantification frameworks to support robust decision-making under geological uncertainty.

The experimental evaluation conducted for each forecasting engine revealed significant variations in predictive model performance, with certain models demonstrating superior accuracy for specific tasks while others proved unsuitable for applications beyond their intended scope (e.g., CRM-type models cannot effectively select candidate wells for workover operations). This finding motivated the development of an integrated workflow incorporating three distinct optimization modules, each designed to address different aspects of reservoir management. The well control optimization module, a key component of this framework, specifically targets the optimization of injection rates for water injectors and bottomhole flowing pressures for production wells. To execute the optimization process, users must first define several critical parameters: (1) the optimization time horizon, which can range from short-term (30-90 days) for operational adjustments to long-term (1-5 years) for strategic planning; (2) physical constraints on control variables, including minimum/maximum injection rates and pressure safety limits; and (3) the optimization scenario that determines the objective function formulation. Available scenario options include:

- Oil production maximization (Net Present Value focus),
- Water injection minimization with production maintenance,
- Operational cost reduction,
- Water cut control,
- Energy efficiency improvement.

The module's architecture (Figure 2) implements a hybrid optimization approach combining gradient-based methods for local search with metaheuristic algorithms (e.g., genetic algorithms) to avoid local optima. Computational efficiency is enhanced through parallelization of simulation cases and implementation of surrogate modeling techniques for rapid objective function evaluation.



$$Q_o = \sum_{s=1}^{N_{prod}} \sum_{k=1}^n q_{oil}^s(t_k) = \sum_{s=1}^{N_{prod}} \sum_{k=1}^n f_o \cdot q_{liq}^s(t_k) \rightarrow \text{maximize}$$

Figure 2—The architecture of the well control parameters optimization module

Field validation studies using history-matched hydrodynamic models demonstrated that optimized control parameters could increase cumulative oil production by 8-12% compared to baseline scenarios that simply extend current operating parameters. Additional benefits included:

- 15-20% reduction in water injection volumes (while maintaining production),
- 5-8% decrease in lifting costs (by mitigating the high level of water cuts),
- Improved sweep efficiency (together with Infill Drilling module) and overall recovery.

This modular approach provides reservoir engineers with a flexible toolkit that can be adapted to various field development stages, from waterflood management in mature fields to production optimization in newly developed assets. The system's architecture allows for continuous expansion with additional optimization modules as new methodologies emerge. Future development directions include implementation of closed-loop optimization with automated parameter updates; incorporation of machine learning for real-time constraint detection; development of multi-objective optimization capabilities; enhanced uncertainty quantification using ensemble-based methods.

Another optimization module is designed for selecting optimal infill drilling locations. Depending on the candidate well's coordinates and assigned operating mode, the algorithm generates a short-term production forecast. Models trained on surrounding wells' historical data are used for this purpose, with their weights interpolated using geostatistical methods (Figure 3) [3]. The methodological pipeline integrates modified CRM (CR-type) modeling with geostatistical interpolation (kriging) to predict production behavior for new wells and update coefficients for existing wells. The approach begins with re-estimating CRM coefficients for existing wells using injection values from the last timestep of the prediction period, which proves particularly effective for reproducing liquid flow rate responses. These re-estimated coefficients then serve as input for kriging interpolation, leveraging spatial correlations and geological data to determine coefficients for new wells. The interpolated coefficients are validated against actual production data (where available) and iteratively refined through variogram adjustments. Finally, the updated coefficients are integrated into the modified CRM to simulate liquid flow rates for new wells and optimize production strategies for existing wells. This hybrid methodology offers an adaptable, physics-informed solution that captures dynamic well interactions while accommodating spatial reservoir characteristics, though it assumes coefficient continuity and requires careful data quality control. The pipeline effectively bridges data-driven modeling with geostatistical approaches, providing a practical tool for reservoir management across different field development stages.

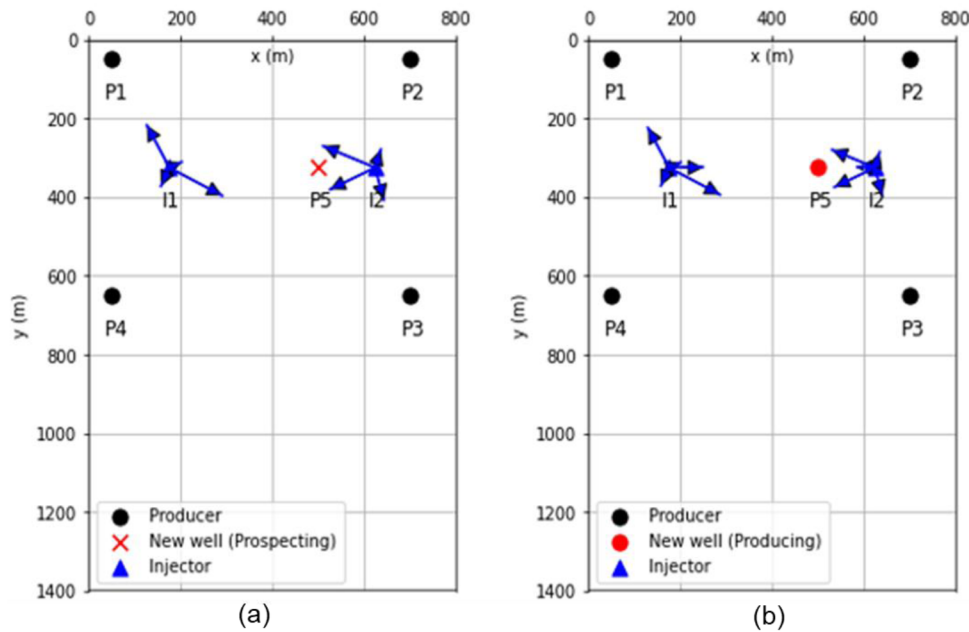


Figure 3—Vectorised maps of Proxy-CRM coefficients: (a) before starting a new well; (b) after starting a new well

Both synthetic and field cases data are used to evaluate the proposed methodology. Experiments showed that the R^2 value for short-term production forecasts of candidate wells can reach 0.9, while for existing wells accounting for new additions, it is 0.7. The optimizer ranks all candidate locations based on cumulative oil production over the selected period. Comparison of metrics obtained for the described cases indicates that tuning the model with selecting clusters of producers within one conduit and considering injectors within radius 3 km from the selected producers without diving them by injection layers is the most suitable way for application of proposed methodology for prediction production behavior after changing the number of producers. After validation it became obvious that

- proxy models such as CRM are capable of capturing the production behaviour of wells using only values of production, injection and BHP without prior knowledge of geological parameters such as permeability, porosity, etc,
- it is essential to tune the model properly and get the most suitable model coefficients in order to perform adequate testing and interpolation procedures,
- kriging allows to obtain coefficients for a new well without data for training that make CR model to give accurate enough predictions for its production behavior.

To advance reservoir management capabilities, we propose developing an integrated universal optimizer that holistically addresses three key operational aspects: well control optimization (BHP/injection regimes), infill drilling strategies, and well treatment campaigns. This AI-driven system will combine physics-based modeling, machine learning, and multi-objective optimization to maximize asset value while respecting operational constraints.

The development of advance reservoir management system based on three steps. The first step is the creation of a unified optimization architecture, including the implementation of hierarchical optimization to prioritize short-term (well control) versus long-term (infill/treatment) decisions, and the development of a modular optimization engine comprising interchangeable submodules for:

- Well control: Adjusting BHP (producers) and injection rates (injectors) via model predictive control (MPC),

- Infill drilling: Identifying optimal locations/timing using surrogate models and uncertainty quantification,
- Well treatments: Scheduling stimulations/workovers based on productivity decline forecasts.

The second step is to train physics-informed neural networks (PINNs) to act as fast surrogates for fluid flow dynamics (coupling CRM with full physics simulations), well interference effects (evaluating the impact of infill drilling) and treatment response prediction (e.g. predicting the success rates of acidising and fracturing). Reinforcement learning (RL) will then be incorporated to enable adaptive strategy updates.

The third step is to implement a multi-objective optimisation algorithm to simultaneously optimise the field development plan (FDP) for: NPV maximisation (oil production versus operational costs) and risk mitigation (water breakthrough, and limits on equipment and infrastructure). For this reason, evolutionary algorithms (e.g. NSGA-III) are well-suited to handling competing objectives.

Conclusions

Three optimizers look for the best schedule for wells, suggest places for infill drilling, and select wells-candidates for workovers, respectively. The best parameters are supposed to be those, which result in a higher cumulative oil production compared to a baseline (planned schedule, events and infill drilling) for the chosen period. As a result, a general optimized field development plan can be obtained.

The novelty of this work lies in integrating different types of models into a single workflow for obtaining a broad field development scenario. As future work, the workflow could be improved by introducing other important aspects of field development, such as predicting well equipment failure for its timely replacement. Also, the addition of near-wellbore flow controlling information [4] can increase the accuracy of the predictive part of the field optimization workflows especially for the reservoirs with complex geology. The target function of FDP optimization can be also enriched with models for forecasting the well treatment effects such ones for hydraulic fracturing presented in [5] and for other types of treatments presented in [6].

Thus, implementing the system that is independent of complex models and is based on machine and deep learning algorithms combined with fundamental physics-driven models, can significantly reduce the need for computational costs of a 3D simulator and routine tasks, if not entirely overhaul the methodology for optimization calculations.

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